The maths behind every MIS report

BUSINESS MATHS

Growth and change

What your manager asks first

Data: Example sheet: B = sales Jan to Jun (1.2 to 1.71 lakh), C = same months last year, F = 7 salaries with one at 2.5 lakh, plus small cells for price, visitors, cost, target.

1 **Month-on-month growth %** → 12.5

```
=ROUND((B3-B2)/B2*100,1)
```

2 **Year-on-year growth %, June** → 14

```
=ROUND((B7-C7)/C7*100,1)
```

3 **Change in rupees, Jan to Jun** → 51,000

```
=B7-B2
```

4 **CAGR: 1 lakh to 1.5 lakh in 3 years** → 14.5

```
=ROUND(((D3/D2)^(1/3)-1)*100,1)
```

5 **Margin change in points** → 5

```
=(E3-E2)*100
```

Say it right: Margin went from 20% to 25%. That is 5 points, not 5%.

BUSINESS MATHS

Averages that lie

Pick the right middle

1 Average monthly sales → 144,333

```
=ROUND(AVERAGE(B2:B7),0)
```

2 Median salary (7 people) → 32,000

```
=MEDIAN(F2:F8)
```

3 Mean salary, same 7 people → 62,857

```
=ROUND(AVERAGE(F2:F8),0)
```

4 Most common order size → 3

```
=MODE(G2:G8)
```

5 Weighted average price → 170

```
=SUMPRODUCT(H2:H4,I2:I4)/SUM(I2:I4)
```

Interview trap: One salary of 2.5 lakh pulls the mean to 62,857. The median stays at 32,000.

BUSINESS MATHS

Spread and outliers

How far apart the numbers are

1 **Standard deviation** → 82,665

```
=ROUND(STDEV(F2:F8),0)
```

2 **90th percentile salary** → 124,000

```
=PERCENTILE(F2:F8,0.9)
```

3 **First quartile (Q1)** → 29,000

```
=QUARTILE(F2:F8,1)
```

4 **IQR (Q3 minus Q1)** → 8,500

```
=QUARTILE(F2:F8,3)-QUARTILE(F2:F8,1)
```

5 **Flag the outlier** → Outlier

```
=IF(F8>QUARTILE($F$2:$F$8,3)+1.5*
(QUARTILE($F$2:$F$8,3)-
QUARTILE($F$2:$F$8,1)),"Outlier","OK")
```

Rule of thumb: Above Q3 + 1.5 x IQR is an outlier. Check it before you delete it.

BUSINESS MATHS

Business ratios

The numbers on every dashboard

1 **Conversion rate** → 4

```
=ROUND(J3/J2*100,1)
```

2 **Profit margin %** → 24

```
=ROUND((K2-K3)/K2*100,1)
```

3 **Average order value** → 200

```
=ROUND(L2/L3,0)
```

4 **June's share of 6-month sales** → 19.7

```
=ROUND(B7/SUM(B2:B7)*100,1)
```

5 **June target achieved %** → 95

```
=ROUND(B7/M2*100,1)
```

Remember: Always say what the % is of. 4% of visitors, not just 4%.

BUSINESS MATHS

Trends and forecasts

Where the numbers are heading

1 Running total to March → 383,000

```
=SUM($B$2:B4)
```

2 3-month moving average → 161,000

```
=ROUND(AVERAGE(B5:B7),0)
```

3 Ad spend vs sales correlation → 0.83

```
=ROUND(CORREL(N2:N7,B2:B7),2)
```

4 Straight-line forecast, July → 180,133

```
=ROUND(FORECAST(7,B2:B7,P2:P7),0)
```

5 How unusual is 2.5 lakh? (z-score) → 2.26

```
=ROUND(STANDARDIZE(F8,AVERAGE(F2:F8),STDEV(F2:F8)),2)
```

Say this: Correlation shows two numbers move together. It does not prove one causes the other.

BUSINESS MATHS

Which number do I report?

You have → Report

1. **Salaries or prices with a few huge values**
   Median, not mean
2. **Prices bought in different quantities**
   Weighted average
3. **Growth over several years**
   CAGR, not total growth divided by years
4. **A change in a % (margin, conversion)**
   Percentage points
5. **A noisy monthly line**
   3-month moving average
6. **One value that looks wrong**
   Check it with IQR or z-score, then ask
7. **Two numbers rising together**
   Correlation, and say it is not proof of cause
8. **Sales vs target**
   Target achieved %, with the target shown

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